

AI for Marketing Decisions

Bridging Machine Learning with Strategic BI Command Architecture

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FINAL EXAM PROJECT

SPECIALTY: AI IN BUSINESS

ASTANA IT UNIVERSITY



Executive Summary & Project Objectives

Transitioning from intuitive marketing to an algorithmic, single-source-of-truth **marketing command center** in Power BI — unifying five pillars of high-performance marketing.



Behavioral Profiling

RFM-based customer classification



Predictive Risk Mitigation

Churn probability via ensemble models



Prescriptive Personalization

Collaborative filtering for cross-selling



Sentiment Intelligence

NLP-driven social media polarity scoring

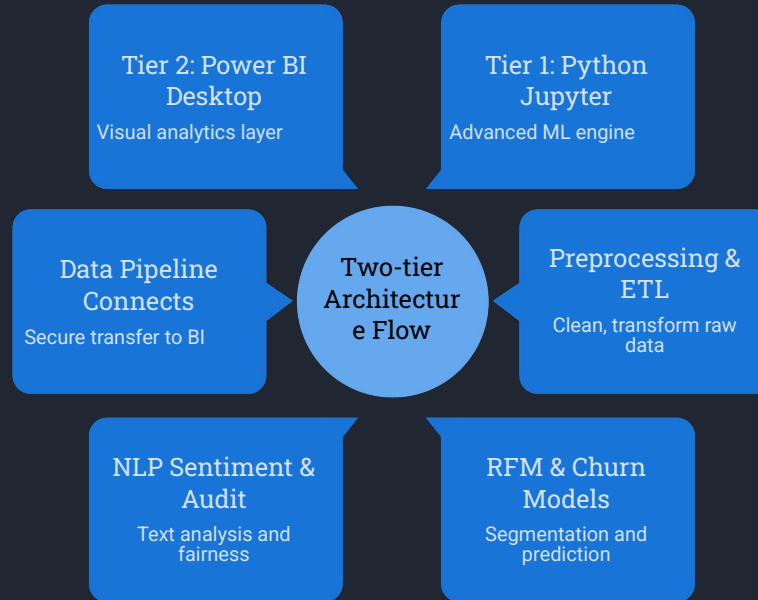


Ethical ML Compliance

Fairness audits and GDPR adherence

Project Architecture & Relational Schema

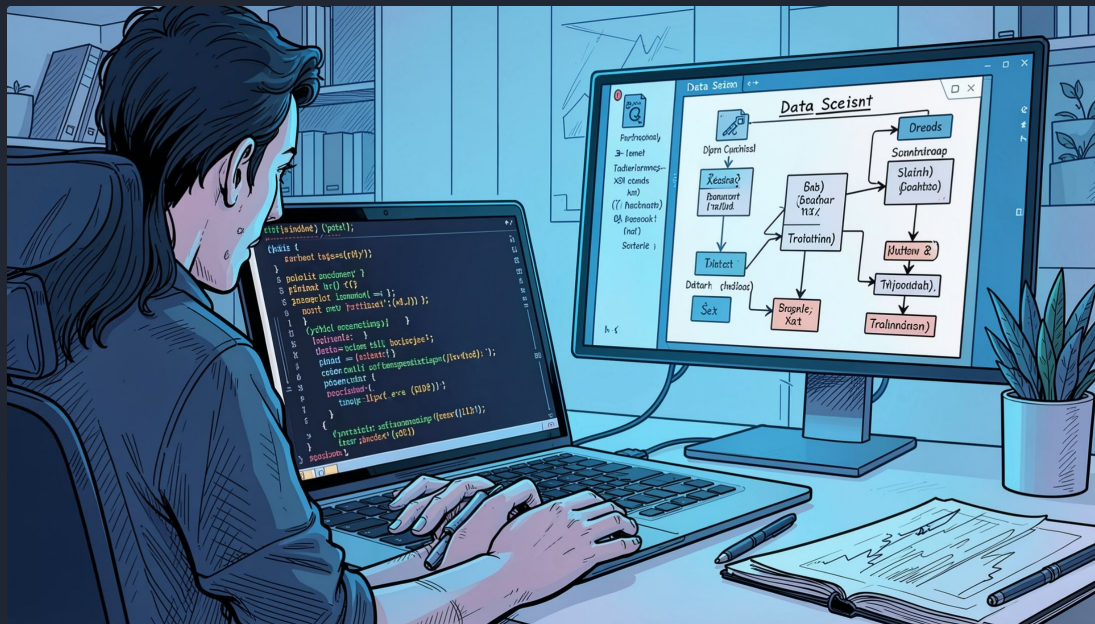
A two-tier architecture connects a Python ML Engine to a Power BI Visual Analytics Layer, with five relational tables forming the data model.



The pipeline flows from raw data processing in Python through to interactive executive dashboards in Power BI.



Advanced Data & Feature Engineering Pipeline



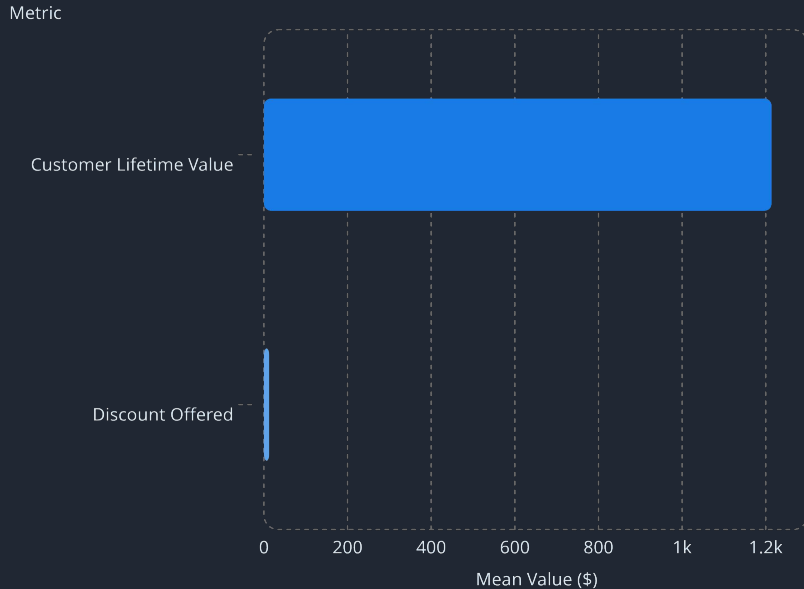
Preprocessing Protocol

- **Median imputation** for missing values — preserving distribution integrity
- **Duplicate removal** across customer transaction logs
- **Data type casting** for modeling compatibility

Engineered Financial Indicators

- **Average Order Value (AOV):** Revenue per transaction — measures purchase intensity
- **Customer Lifetime Value (CLV):** Cumulative revenue per account — identifies long-term

Exploratory Data Analysis & Key Distributions



CLV Distribution – Right-Skewed

Mean CLV of **\$1,214.4** with a right-skewed profile confirms that a small cohort of repeat buyers drives the majority of corporate revenue – validating VIP retention investment.

Discount Distribution – Multi-Modal

Mean discount of **\$13.4** with a multi-modal pattern confirms distinct promotional campaign tiers operating simultaneously – enabling segment-specific offer optimization.

Unsupervised Learning: RFM Behavioral Segmentation

RFM vectors replace rigid demographic buckets, classifying customers into three operationally distinct tiers for targeted intervention.

1

High-Value Loyalists

Recent, frequent, high-spend — retain via **exclusive VIP programs** and early access

2

At-Risk Core Spenders

Declining recency with strong history — trigger **automated win-back offers**

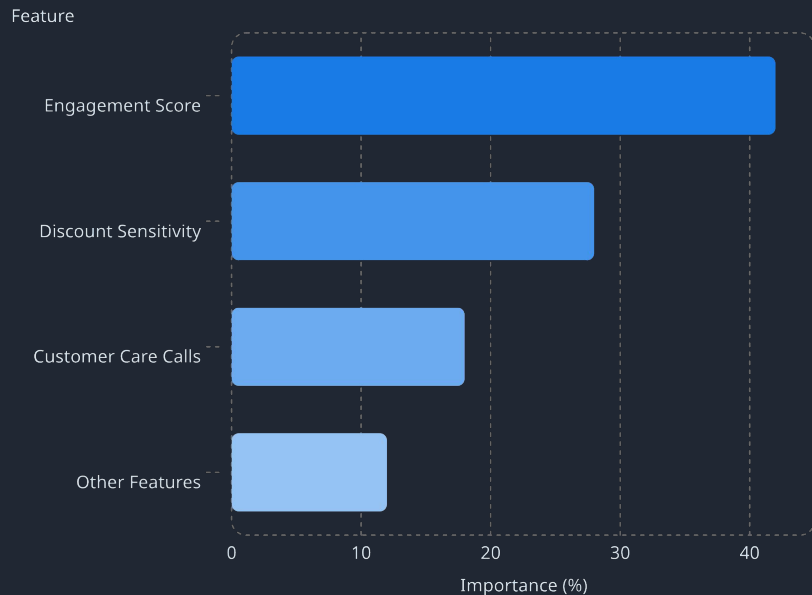
3

Bargain Hunters

High discount sensitivity, low frequency — deploy **margin-optimized flash sales**



Supervised Learning: Predictive Churn Risk Modeling



Ensemble Classification Model

Random Forest and XGBoost ensemble outputs **HVC_Probability** — a continuous churn risk score per customer account, evaluated via Confusion Matrix for precision and recall.

Top Predictive Drivers

- **Engagement Score (42%):** Dominant predictor of retention intent
- **Discount Sensitivity (28%):** Price dependency as churn signal
- **Customer Care Calls (18%):** Service friction as early-warning indicator

Prescriptive Personalization & Sentiment Intelligence

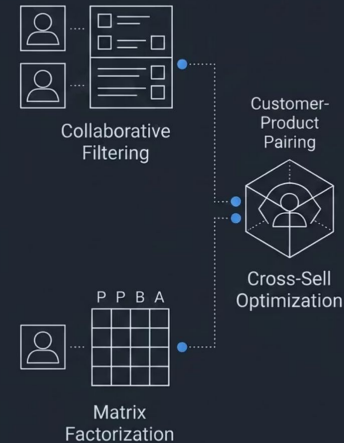
Product Recommendation Engine

Collaborative filtering via **matrix factorization** dynamically pairs each Customer ID with top-predicted product categories — maximizing cross-sell potential and average basket size without manual rules.

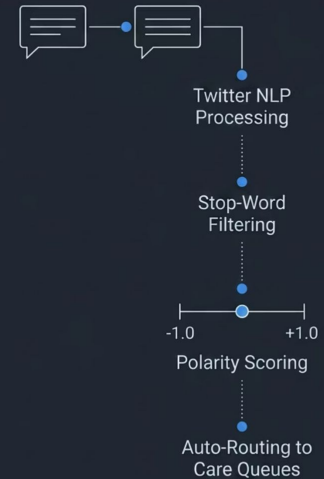
NLP Sentiment Intelligence

Unstructured Twitter text is processed through stop-word filtering and tokenization, generating continuous **Polarity Scores (-1.0 to +1.0)**. Accounts exceeding negative thresholds are auto-routed to customer care queues for proactive intervention.

Recommendation Engine



Sentiment Intelligence





DIGITAL COMPLIANCES

Algorithmic Governance & Responsible AI Compliance

Algorithmic Fairness Audit

The **Disparate Impact Ratio (DIR)** is tracked across protected demographics within the `fairness_summary` layer. Compliance threshold: **DIR between 0.80 and 1.25** — ensuring no segment experiences systematically biased model outcomes.

GDPR Data Compliance

All **Personally Identifiable Information (PII)** undergoes complete anonymization and cryptographic hashing at the database source layer — prior to any ML processing — ensuring full regulatory adherence and auditability.

Strategic Recommendations & Implementation Roadmap

01

Phase 1 — Pipeline Cleansing (Days 1–30)

Validate data quality; deploy PII hashing

02

Phase 2 — Pilot Testing (Days 31–60)

Launch churn model and RFM segmentation on live data

03

Phase 3 — Full Automation (Days 61–90)

Automate interventions, recommendations, and sentiment routing

Actionable Takeaways

- Trigger automated risk interventions for high-churn accounts
- Inject personalized affinity mappings into marketing channels
- Optimize margins for price-inelastic loyalist segments
- Establish sentiment-driven support routing workflows



Project Portfolio: Access full source code, preprocessed datasets, and complete technical report at:

<https://github.com/ilyas06089-collab/Ilyas>